

EE3621: Digital Signal Processing

Lecture 29: IIR filter design - matched z-transform; simple design example (III B.Tech EEE)

September 23, 2026

1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Apply** the Matched Z-Transform to convert analog transfer functions to discrete transfer functions.
 2. **Design** simple digital IIR filters using the Matched Z-Transform method.
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2. MATHEMATICAL FOUNDATIONS

2.1 The Matched Z-Transform (MZT)

Given analog factored transfer function:

$$H_a(s) = K \frac{\prod_{i=1}^M (s - z_i)}{\prod_{k=1}^N (s - p_k)}$$

The discrete transfer function is formed by mapping each factor:

$$H(z) = K_d \frac{\prod_{i=1}^M (1 - e^{z_i T_d} z^{-1})}{\prod_{k=1}^N (1 - e^{p_k T_d} z^{-1})} (1 + z^{-1})^{N-M}$$

Where K_d is chosen to match the DC gain or passband peak.

Pictorial Representations

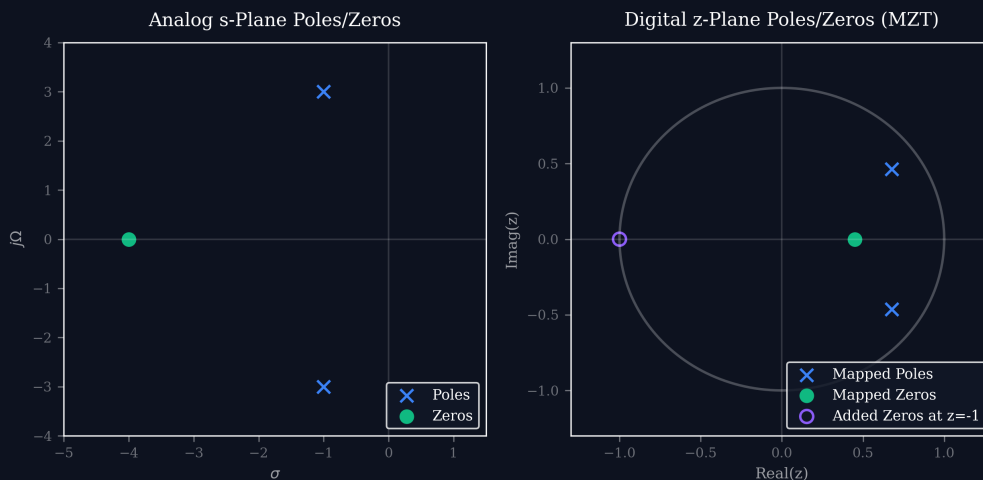


Figure 1: Matched Zplane

3. WORKED NUMERICAL EXAMPLES

Example 29.1: Simple Matched Z-Transform Design

Problem: Design a digital IIR lowpass filter using the Matched Z-Transform method given the analog transfer function:

$$H_a(s) = \frac{1}{s + 2}$$

Assume a sampling frequency of $F_s = 10$ Hz (or sampling period $T_d = 0.1$ s). Set the DC gain of the digital filter to match the DC gain of the analog filter.

Solution:

1. **Identify poles and zeros of $H_a(s)$:** The analog transfer function has no finite zeros ($M = 0$) and one pole at $s_1 = -2$ ($N = 1$).
2. **Map poles and zeros to the z-plane:** The single analog pole maps to a digital pole:

$$z_1 = e^{s_1 T_d} = e^{-2 \times 0.1} = e^{-0.2} \approx 0.8187$$

3. **Formulate the unscaled digital transfer function:** According to the MZT formulation, we map finite poles and zeros and add $(1 + z^{-1})^{N-M} = (1 + z^{-1})^1$ to the numerator to map the analog zero at infinity to $z = -1$ (the Nyquist frequency):

$$H(z) = K_d \frac{1 + z^{-1}}{1 - e^{-0.2} z^{-1}} = K_d \frac{1 + z^{-1}}{1 - 0.8187 z^{-1}}$$

4. **Determine the gain factor K_d :** Match the DC gain ($s = 0$ or $\omega = 0$) of the analog filter:

$$H_a(0) = \frac{1}{0 + 2} = 0.5$$

Evaluate the unscaled digital filter at DC ($z = 1$):

$$H(1) = K_d \frac{1 + 1}{1 - 0.8187} = K_d \frac{2}{0.1813} = 11.0314 K_d$$

Equating the two:

$$11.0314K_d = 0.5 \implies K_d \approx 0.0453$$

5. Final Digital Transfer Function:

$$H(z) = \frac{0.0453(1 + z^{-1})}{1 - 0.8187z^{-1}}$$